Discrete Random Variables and Distributions

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Random Variables

In probability theory, a **random variable** (RV) X is a function of outcomes.

Recall

- Ω is a sample space of outcomes.
- ω denotes some specified outcome.
- $X(\omega)$ is a real value for every $\omega \in \Omega$.
- $P(X(\omega) = x)$ is the probability $X(\omega) = x$.

We only consider discrete RVs in this chapter, namely the set of possible values of $X(\omega)$ is finite or countable.

A tease of discrete versus continuous RVs

Discrete RV examples:

- Number of Heads by tossing a coin n times.
- Number of tosses to get the first Head.
- Number of files infected by virus.

Continuous RV examples:

- Software installation time.
- Lifetime of a bulb.
- A real number randomly drawn from an interval.

Example

Toss a coin 3 times and let X denote the number of heads. Find Ω with all possible outcomes ω and values of $X(\omega)$.

Solution

We denote Head and Tail by H and T. Tossing the coin 3 times may have 8 outcomes:

$$\text{(T,T,T), (T,T,H), (T,H,T), (T,H,H),}$$
$$\text{(H,T,T), (H,T,H), (H,H,T), (H,H,H).}$$

- Ω is the set of these 8 outcomes.
- ω can be any of these 8 outcomes.
- If $\omega =$ (T, H, T), then $X(\omega) = 1$, and so on.
- 0, 1, 2, and 3 are all possible values of $X(\omega)$.

Distribution Functions

We often omit ω because X is often clear from the context. Note $X = x$ may be a set of many outcomes. For the example above, $X = 2$ is the set {(H,H,T), (H,T,H),(T,H,H)}.

We call p the **probability mass function** (pmf) of X if

$$p(x) = P(X = x)$$

for every possible value x of X.

We call F the **cumulative distribution function** (cdf) of X if

$$F(x) = P(X \leq x)$$

for any $x \in \mathbb{R}$.

Properties of pmf

- $p(x) \geq 0$.
- $p(x) > 0$ if and only if x is a possible value of X.
- $\sum_x p(x) = 1$ (the sum is over all possible values of X).

Properties of cdf

- F is non-decreasing.
- (Discrete RV only) F is staircase-like and only jumps at possible values of X.
- (Discrete RV only) $F(x) = P(X \leq x) = \sum_{k \leq x} P(X = k)$.
- $0 \leq F(x) \leq 1$ for all $x \in \mathbb{R}$.
- The trend of F:

$$\lim_{x \to -\infty} F(x) = 0, \qquad \lim_{x \to +\infty} F(x) = 1$$

Example

Toss a **fair** coin 3 times. Let X be the number of heads. Find pmf p and cdf F.

Solution

- $X = 0$ means {(T,T,T)}, hence $p(0) = \frac{1}{8}$.
- $X = 1$ means {(T,T,H),(T,H,T),(H,T,T)}, hence $p(1) = \frac{3}{8}$.
- Similarly $p(2) = \frac{3}{8}$, $p(3) = \frac{1}{8}$.

The pmf p is

x	0	1	2	3
$p(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Solution (cont)

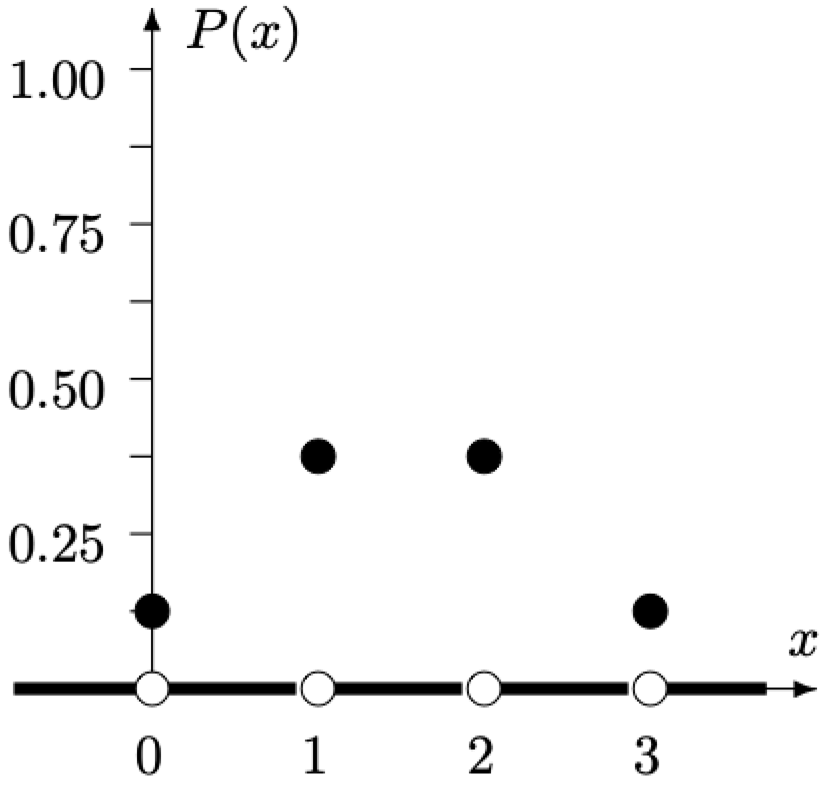
The cdf F is

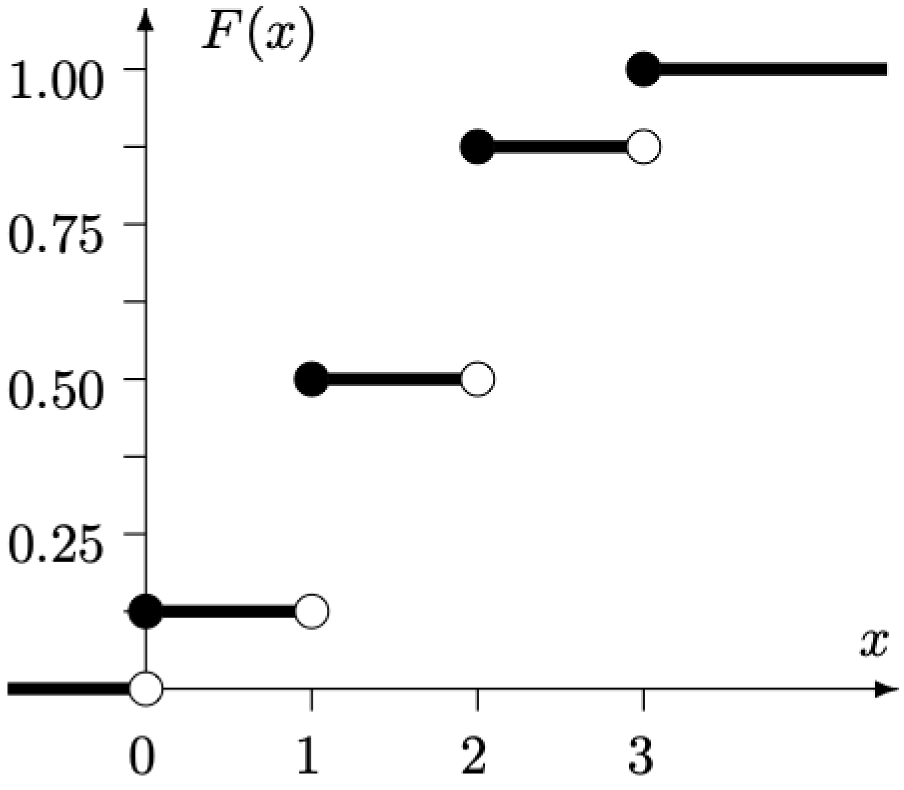
$$F(x) = \begin{cases} 0, & \text{if } x < 0, \\ \frac{1}{8}, & \text{if } 0 \leq x < 1, \\ \frac{4}{8}, & \text{if } 1 \leq x < 2, \\ \frac{7}{8}, & \text{if } 2 \leq x < 3, \\ 1, & \text{if } x \geq 3. \end{cases}$$

Remark: The cdf of any discrete RV is a non-decreasing staircase function like this.

Solution (cont)

Plots of pmf p and cdf F:





Example. Let X and Y be two independent RVs with pmfs p_X and p_Y:

k	0	1	2	3
$p_X(k)$	0.5	0.3	0.1	0.1
$p_Y(k)$	0.7	0.2	0.1	0

Find the pmf and cdf of $Z = X + Y$.

Solution. We see Z can take any integer between 0 and 5.

For the pmf p_Z, we see for example

$$Z = X + Y = 1 \quad \iff \quad \{(X = 1, Y = 0), (X = 0, Y = 1)\}$$

and thus

$$\begin{aligned} p_Z(1) = P(Z = 1) &= P(X = 1, Y = 0) + P(X = 0, Y = 1) \\ &= P(X = 1)P(Y = 0) + P(X = 0)P(Y = 1) \\ &= 0.31 \end{aligned}$$

We can compute p_Z at other values similarly.

Solution (cont) The pmf of Z is

k	0	1	2	3	4	5
$p_Z(k)$	0.35	0.31	0.18	0.12	0.03	0.01

The cdf of Z is

$$F_Z(x) = \begin{cases} 0, & \text{if } x < 0, \\ 0.35, & \text{if } 0 \le x < 1, \\ 0.66, & \text{if } 1 \le x < 2, \\ 0.84, & \text{if } 2 \le x < 3, \\ 0.96, & \text{if } 3 \le x < 4, \\ 0.99, & \text{if } 4 \le x < 5, \\ 1, & \text{if } x \ge 5. \end{cases}$$

Joint distribution

In real-life, the RVs of interest are usually **dependent**. How they behave together?

We call $p_{X,Y}$ the **joint distribution** of X and Y if

$$p_{X,Y}(x, y) = P(X = x \text{ and } Y = y)$$

for any x, y.

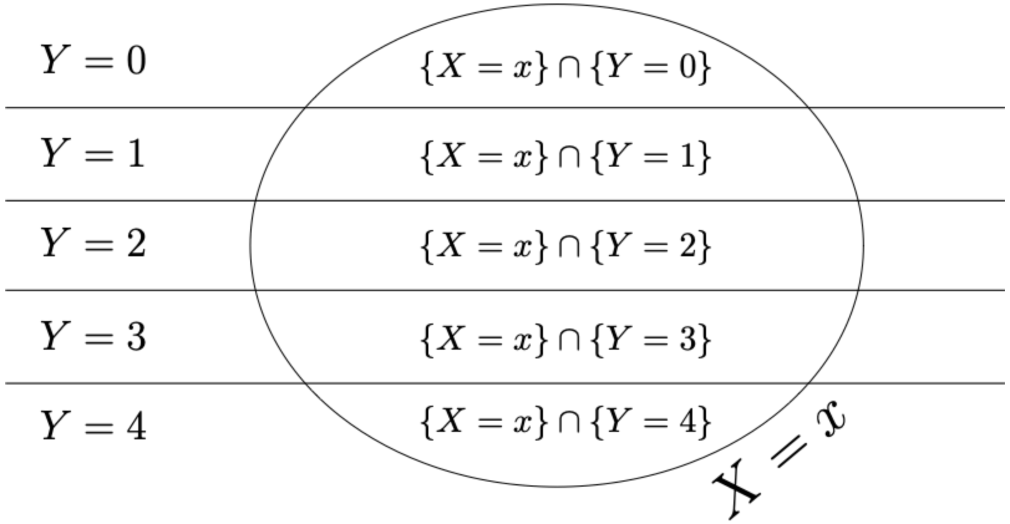
Joint distribution and marginal distribution

Note the relation between pmf p_X, p_Y and $p_{X,Y}$:

$$p_X(x) = P(X = x) = \sum_y P(X = x \text{ and } Y = y) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = P(Y = y) = \sum_x P(X = x \text{ and } Y = y) = \sum_x p_{X,Y}(x, y)$$

We also call p_X and p_Y **marginal distribution** in this context.



We call X and Y **independent** if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y), \qquad \forall\, x, y.$$

Expectation (aka Expected Value, Mean, Average)

Example. Consider tossing a fair coin. Win \$2 if Head. Lose \$3 if Tail. What is the expected gain per toss?

Intuition

Imagine we toss it for n times (n very large), we expect both Heads and Tails appear $\frac{n}{2}$ times. So the total gain is

$$2 \times \frac{n}{2} + (-3) \times \frac{n}{2} = -\frac{n}{2}.$$

So the expected gain per toss is the total gain divided n, which is $-\frac{1}{2}$. This is the expectation.

Expectation (aka Expected Value, Mean, Average)

Definition. The **expectation** of a discrete RV X is defined by

$$\mathbb{E}(X) = \sum_x xp(x)$$

where the sum is over all possible values of X.

- Shorthand notation: μ_X, μ.
- Expectation is a **weighted sum** of all possible values of X.
- The weight of x is $p(x)$.

Example. Find $\mathbb{E}(X)$ of X whose pmf is

x	0	1	2	3
$p(x)$	0.5	0.3	0.1	0.1

Solution

$$
\begin{aligned}
\mathbb{E}(X) = \sum_{x=0}^{3} xp(x) &= 0 \cdot p(0) + 1 \cdot p(1) + 2 \cdot p(2) + 3 \cdot p(3) \\
&= 0 \cdot 0.5 + 1 \cdot 0.3 + 2 \cdot 0.1 + 3 \cdot 0.1 \\
&= 0.8.
\end{aligned}
$$

Expectation of a function

Let X be a RV and $Y = g(X)$ for some function g. Then

$$\mathbb{E}(Y) = \mathbb{E}(g(X)) = \sum_x g(x)p(x)$$

- Obvious when g is one-to-one.
- In general, $y = g(x)$ gets the same weight $p(x)$ as x does.

Linearity of expectation. For any RVs X and Y, and $a, b, c \in \mathbb{R}$, there is

$$\mathbb{E}(aX + bY + c) = a\mathbb{E}(X) + b\mathbb{E}(Y) + c$$

Proof

$$\begin{aligned}
\mathbb{E}(aX + bY + c) &= \sum_x \sum_y (ax + by + c) P(X = x, Y = y) \\
&= a \sum_x \sum_y x P(X = x, Y = y) + b \sum_x \sum_y P(X = x, Y = y) + c \sum_x \sum_y P(X = x, Y = y) \\
&= a \sum_x x \Big(\sum_y P(X = x, Y = y) \Big) + b \sum_y y \Big(\sum_x P(X = x, Y = y) \Big) + c \\
&= a \sum_x x P(X = x) + b \sum_y y P(Y = y) + c \\
&= a\mathbb{E}(X) + b\mathbb{E}(Y) + c.
\end{aligned}$$

Expectation of product

If X and Y are independent, then

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$$

Proof

$$\begin{aligned}\mathbb{E}(XY) = \sum_x \sum_y (xy)P(X = x, Y = y) = \sum_x \sum_y (xy)P(X = x)P(Y = y) \\ = \Big(\sum_x xP(X = x)\Big) \cdot \Big(\sum_y yP(Y = y)\Big) = \mathbb{E}(X)\mathbb{E}(Y).\end{aligned}$$

Warning: The converse is not necessarily true!

Measure of uncertainty

The **variance** of X is defined by

$$\mathrm{Var}(X) = \mathbb{E}\big((X-\mu)^2\big) = \sum_x (x-\mu)^2 p(x)$$

Shorthand notations of variance: σ_X^2, σ^2.

The **standard deviation** of X is

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

Sometimes also written as $\mathrm{std}(X)$.

Remark. $\mathrm{Var}(X)$ is always non-negative.

Claim

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - \mu^2$$

Proof

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}((X-\mu)^2) = \sum_x (x-\mu)^2 p(x) \\
&= \sum_x (x^2 - 2\mu x + \mu^2) p(x) \\
&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\
&= \mathbb{E}(X^2) - 2\mu^2 + \mu^2 \\
&= \mathbb{E}(X^2) - \mu^2
\end{aligned}$$

Claim

$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$$

Proof

$$\mathbb{E}\big((aX+b)^2\big) = \mathbb{E}(a^2X^2 + 2abX + b^2) = a^2\mathbb{E}(X^2) + 2ab\mu + b^2$$
$$\big(\mathbb{E}(aX+b)\big)^2 = (a\mu+b)^2 = a^2\mu^2 + 2ab\mu + b^2$$

Subtraction gives

$$\begin{aligned}\mathrm{Var}(aX+b) &= \mathbb{E}\big((aX+b)^2\big) - \big(\mathbb{E}(aX+b)\big) \\ &= a^2\mathbb{E}(X^2) - a^2\mu^2 \\ &= a^2\mathrm{Var}(X)\end{aligned}$$

Remark. b is gone because it is an added constant and has no randomness.

Covariance between two RVs

The **covariance** of X and Y is defined by

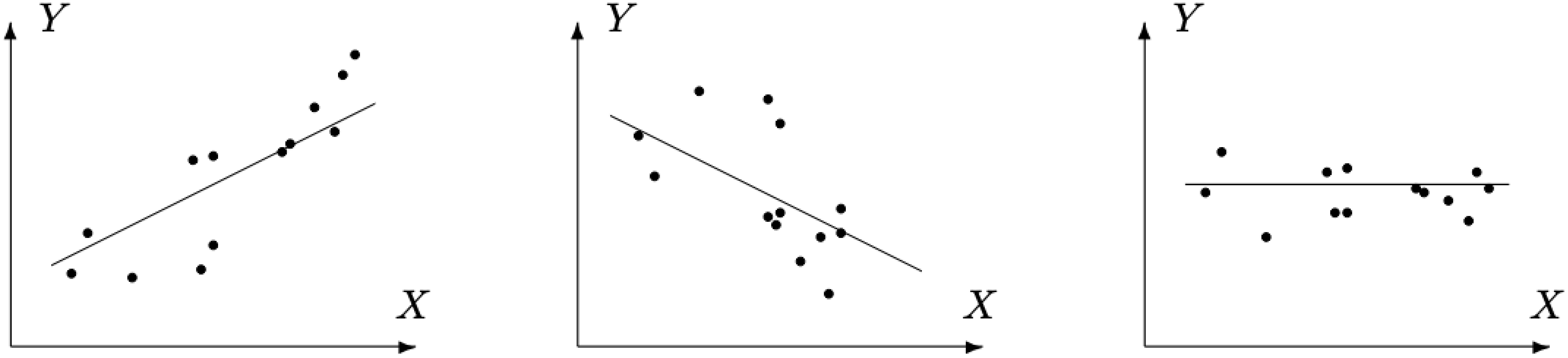
$$\mathrm{Cov}(X, Y) = \mathbb{E}\big((X - \mu_X)(Y - \mu_Y)\big)$$

We also see:

$$\begin{aligned}
\mathrm{Cov}(X, Y) &= \mathbb{E}\big((X - \mu_X)(Y - \mu_Y)\big) \\
&= \mathbb{E}\big(XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y\big) \\
&= \mathbb{E}(XY) - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\
&= \mathbb{E}(XY) - \mu_X \mu_Y
\end{aligned}$$

Important! If X and Y are independent, then $\mathbb{E}(XY) = \mu_X \mu_Y$ and $\mathrm{Cov}(X, Y) = 0$. **However,** $\mathrm{Cov}(X, Y) = 0$ **does not necessarily imply** X **and** Y **are independent**.

$\mathrm{Cov}(X, Y) > 0$ implies "larger X suggests larger Y" and etc.



(a) $\mathrm{Cov}(X, Y) > 0$ (b) $\mathrm{Cov}(X, Y) < 0$ (c) $\mathrm{Cov}(X, Y) = 0$

Properties of covariance

- $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
- $\mathrm{Cov}(aX + b, Y) = a\mathrm{Cov}(X, Y)$

Proof. Try first two yourself. For the last one, there is

$$\begin{aligned}
\mathrm{Cov}(aX + b, Y) &= \mathbb{E}\big((aX + b)Y\big) - \mathbb{E}(aX + b)\mathbb{E}(Y) \\
&= \mathbb{E}(aXY + bY) - (a\mu_X + b)\mu_Y \\
&= a\mathbb{E}(XY) + b\mu_Y - a\mu_X\mu_Y - b\mu_Y \\
&= a\mathbb{E}(XY) - a\mu_X\mu_Y \\
&= a\mathrm{Cov}(X, Y)
\end{aligned}$$

Claim

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X,Y) + \mathrm{Var}(Y)$$

Proof. We have

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathbb{E}\big((X+Y)^2\big) - \big(\mathbb{E}(X+Y)\big)^2 \\
&= \mathbb{E}(X^2 + 2XY + Y^2) - (\mu_X + \mu_Y)^2 \\
&= \mathbb{E}(X^2) + 2\mathbb{E}(XY) + \mathbb{E}(Y^2) - \mu_X^2 - 2\mu_X\mu_Y - \mu_Y^2 \\
&= \mathrm{Var}(X) + 2\mathrm{Cov}(X,Y) + \mathrm{Var}(Y)
\end{aligned}$$

Important special case: If X and Y are independent, then

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

Correlation between two RVs

The **correlation** between X and Y is defined by

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Shorthand notation: $\rho_{X,Y}$ or ρ.
- Correlation is a **normalization** of covariance.

Claim

$$-1 \leq \mathrm{Corr}(X,Y) \leq 1$$

Proof. Define $U = \frac{X-\mu}{\sigma_X}$ and $V = \frac{Y-\mu_Y}{\sigma_Y}$. Then

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \mathrm{Cov}\left(\frac{X-\mu_X}{\sigma_X}, \frac{Y-\mu_Y}{\sigma_Y}\right) = \mathrm{Cov}(U,V).$$

Notice $\mu_U = \mu_V = 0$ and $\mathbb{E}(U^2) = \mathbb{E}(V^2) = 1$, by Cauchy-Schwarz inequality,

$$|\mathrm{Cov}(U,V)| = |\mathbb{E}(UV) - \mu_U \mu_V| = |\mathbb{E}(UV)| \leq \sqrt{\mathbb{E}(U^2)}\sqrt{\mathbb{E}(V^2)} = 1$$

Combining the two above yields the claimed bounds.

Remark. $\mathrm{Corr}(X,Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$.

Chebyshev's inequality

Let X be any RV with expectation μ and variance σ^2, and $\epsilon > 0$ arbitrary. Then

$$P(|X-\mu| > \epsilon) \leq \left(\frac{\sigma}{\epsilon}\right)^2$$

Proof.

$$\begin{aligned}\sigma^2 &= \sum_x (x-\mu)^2 p(x) \geq \sum_{\{x:|x-\mu|>\epsilon\}} p(x) \\ &= \sum_{\{x:|x-\mu|>\epsilon\}} \epsilon^2 p(x) = \epsilon^2 \sum_{\{x:|x-\mu|>\epsilon\}} p(x) = \epsilon^2 P(|X-\mu| > \epsilon)\end{aligned}$$

Remark. Provides a fast (but often rough) estimate of the deviation $|X-\mu|$.

Example. Suppose a RV X has expectation 20 and standard deviation 2. Use the Chebyshev's inequality to upper bound $P(X > 30)$.

Solution. We have

$$P(X > 30) \leq P(|X - 20| > 10) \leq \left(\frac{2}{20}\right)^2 = 0.04$$

Bernoulli distribution

Let $\theta \in (0, 1)$. We say $X \sim \mathrm{Bernoulli}(\theta)$ if its pmf is

$$p(x) = \begin{cases} \theta, & \text{if } x = 1, \\ 1 - \theta, & \text{if } x = 0. \end{cases}$$

We find

$$\begin{aligned} \mathbb{E}(X) &= 1 \cdot \theta + 0 \cdot (1 - \theta) = \theta \\ \mathbb{E}(X^2) &= 1^2 \cdot \theta + 0^2 \cdot (1 - \theta) = \theta \\ \mathrm{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \theta - \theta^2 = \theta(1 - \theta) \end{aligned}$$

Example. Tossing a coin with Head chance θ, with $X = 1$ for Head and $X = 0$ for Tail.

Binomial distribution

Toss a coin with Head chance $\theta \in (0, 1)$ for n times, and let X be the number of Heads.

We say $X \sim \mathrm{Binomial}(n, \theta)$ with pmf

$$p(x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$$

for $x = 0, 1, \ldots, n$.

Remarks.

- We can think $X = Y_1 + \cdots + Y_n$, where $Y_i \sim \mathrm{Bernoulli}(\theta)$ and Y_i's are independent.
- If $X \sim \mathrm{Binomial}(n, \theta)$, then $(n - X) \sim \mathrm{Binomial}(n, 1 - \theta)$.

Recall Bernoulli(θ) has expectation θ and variance $\theta(1-\theta)$.

Hence for $X \sim \mathrm{Binomial}(n, \theta)$, we have

$$\mathbb{E}(X) = \mathbb{E}\Big(\sum_{i=1}^{n} Y_i\Big) = \sum_{i=1}^{n} \mathbb{E}(Y_i) = n\theta$$

$$\mathrm{Var}(X) = \mathrm{Var}\Big(\sum_{i=1}^{n} Y_i\Big) = \sum_{i=1}^{n} \mathrm{Var}(Y_i) = n\theta(1-\theta)$$

where splitting the variance of $\sum_{i=1}^{n} Y_i$ is due to their independence.

Example. Roll a fair die for 5 times. Let X be the number of times that 6 is rolled. Find $\mathbb{E}(X)$, $\mathrm{Var}(X)$, and the probability of rolling 6 for more than once.

Solution. Notice $X \sim \mathrm{Binomial}(5, \frac{1}{6})$. We have

$$\mathbb{E}(X) = n\theta = \frac{5}{6}, \qquad \mathrm{Var}(X) = n\theta(1-\theta) = \frac{25}{36}.$$

and

$$\begin{aligned} P(X > 1) &= 1 - P(X = 0) - P(X = 0) \\ &= 1 - \binom{n}{0}\theta^0(1-\theta)^{n-0} - \binom{n}{1}\theta^1(1-\theta)^{n-1} \\ &= 1 - \binom{5}{0}\left(\frac{1}{6}\right)^0\left(1 - \frac{1}{6}\right)^{5-0} - \binom{5}{1}\left(\frac{1}{6}\right)^1\left(1 - \frac{1}{6}\right)^{5-1} \\ &\approx 0.1962 \end{aligned}$$

Geometric distribution

Toss a coin with Head chance $\theta \in (0,1)$ until the first Head appears. Let X be the number of tosses.

We say $X \sim \mathrm{Geometric}(\theta)$ if its pmf is

$$p(x) = (1-\theta)^{x-1}\theta$$

for $x = 1, 2, \ldots$

Remark. We get $x - 1$ Tails and then a Head. Hence the pmf p here.

Some useful identities

Let $r \in (0,1)$. Then

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$$

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2}$$

$$\sum_{k=2}^{\infty} k(k-1) r^{k-2} = \frac{2}{(1-r)^3}$$

Proof.

1. Let $S = \sum_{k=0}^{\infty} r_k$. Then $S - rS = 1$ and hence $S = \frac{1}{1-r}$.
2. Notice

$$\sum_{k=1}^{\infty} kr^{k-1} = \sum_{k=1}^{\infty} \frac{d}{dr}(r^k) = \frac{d}{dr}\left(\sum_{k=1}^{\infty} r^k\right) = \frac{d}{dr}\left(\frac{r}{1-r}\right) = \frac{1}{(1-r)^2}$$

3. Similarly,

$$\sum_{k=2}^{\infty} k(k-1)r^{k-2} = \sum_{k=2}^{\infty} \frac{d^2}{dr^2}(r^k) = \frac{d^2}{dr^2}\left(\sum_{k=2}^{\infty} r^k\right) = \frac{d^2}{dr^2}\left(\frac{r^2}{1-r}\right) = \frac{2}{(1-r)^3}$$

Let $r = 1 - \theta$ and $k = x$, we get

$$\mathbb{E}(X) = \sum_{x=1}^{\infty} xp(x) = \sum_{x=1}^{\infty} x(1-\theta)^{x-1}\theta = \frac{\theta}{(1-(1-\theta))^2} = \frac{1}{\theta}$$

$$\mathbb{E}(X^2 - X) = \sum_{k=2}^{\infty}(x^2 - x)(1-\theta)^{x-1}\theta = \frac{2(1-\theta)\theta}{(1-(1-\theta))^3} = \frac{2(1-\theta)}{\theta^2}$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2 - X) + \mathbb{E}(X) - (\mathbb{E}(X))^2 = \frac{2(1-\theta)}{\theta^2} + \frac{1}{\theta} - \frac{1}{\theta^2} = \frac{1-\theta}{\theta^2}$$

Example. Roll a fair die until the first 3 appears. Let X be the number of rolls. Find the pmf p, $\mathbb{E}(X)$, and $\mathrm{Var}(X)$.

Solution. Notice $X \sim \mathrm{Geometric}(\frac{1}{6})$. We have its pmf

$$p(x) = (1-\theta)^{x-1}\theta = \left(\frac{5}{6}\right)^{x-1}\frac{1}{6}$$

for $x = 1, 2, \ldots$ Moreover

$$\mathbb{E}(X) = \frac{1}{\theta} = \frac{1}{\frac{1}{6}} = 6$$

$$\mathrm{Var}(X) = \frac{1-\theta}{\theta^2} = \frac{1-\frac{1}{6}}{\left(\frac{1}{6}\right)^2} = 30$$

Negative Binomial distribution

Toss a coin with Head chance $\theta \in (0,1)$ until the k-th Head appears. Let X be the number of tosses.

We say $X \sim \mathrm{NB}(k, \theta)$ if it has pmf

$$p(x) = \binom{x-1}{k-1}(1-\theta)^{x-k}\theta^k$$

for $x = k, k+1, \ldots$

Remark. We get $(k-1)$ Heads and $(x-k)$ Tails in the first $(x-1)$ tosses, and then a Head in the x-th toss. Hence the pmf p here.

Remark. We can think $X = Y_1 + \cdots + Y_k$, where $Y_i \sim \mathrm{Geometric}(\theta)$ and Y_i's are independent.

Recall $\mathrm{Geometric}(\theta)$ has expectation $\frac{1}{\theta}$ and variance $\frac{1-\theta}{\theta^2}$.

Hence for $X \sim \mathrm{NB}(k, \theta)$, we have

$$\mathbb{E}(X) = \mathbb{E}\Big(\sum_{i=1}^{k} Y_i\Big) = \sum_{i=1}^{k} \mathbb{E}(Y_i) = \frac{k}{\theta}$$

$$\mathrm{Var}(X) = \mathrm{Var}\Big(\sum_{i=1}^{k} Y_i\Big) = \sum_{i=1}^{k} \mathrm{Var}(Y_i) = \frac{k(1-\theta)}{\theta^2}$$

where splitting the variance of $\sum_{i=1}^{n} Y_i$ is due to their independence.

Example. A certain electronic component has 5% chance to be defective. Let X be the number of components we check until 12 non-defective ones are found. Find the probability that $X > 15$.

Solution. Notice $X \sim \mathrm{NB}(12, 0.95)$. We have its pmf

$$p(x) = \binom{x-1}{k-1}(1-\theta)^{x-k}\theta^{12} = \binom{x-1}{11}(0.05)^{x-12}(0.95)^{12}$$

for $x = 12, 13, \ldots$ Hence

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{x=12}^{15} p(x)$$

Poisson distribution

Fix a time period during which a certain event independently occurs $\lambda > 0$ times on average. Let X be the number of actual occurrences.

We say $X \sim \mathrm{Poisson}(\lambda)$ it has pmf

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

for $x = 0, 1, \ldots$

Remark. The unity of p is due to the Taylor expansion

$$e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

We have

$$\mathbb{E}(X) = \sum_{x=0}^{\infty} xp(x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}\lambda^x}{x!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} = \lambda$$

and

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}(X^2 - X) + \mathbb{E}(X) - (\mathbb{E}(X))^2 \\
&= \sum_{x=0}^{\infty} x(x-1) \cdot \frac{e^{-\lambda}\lambda^x}{x!} + \lambda - \lambda^2 \\
&= \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda - \lambda^2 \\
&= \lambda
\end{aligned}$$

Example. An Internet service provider opens 10 new accounts per day on average. Let X be the number of new accounts opened in 2 days. Find $P(X > 16)$.

Solution. Notice $X \sim \mathrm{Poisson}(20)$. We have its pmf

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-20}20^x}{x!}$$

for $x = 0, 1, 2, \ldots$ Hence

$$P(X > 16) = 1 - P(X \leq 16) = 1 - \sum_{x=0}^{16} p(x)$$

Claim. Let $\lambda > 0$ and n be very large, and $\theta = \frac{\lambda}{n} \in (0,1)$ which is close to 0, then

$$\text{Binomial}(n, \theta) \approx \text{Poisson}(n\theta)$$

in the sense that for any $x = 0, 1, \ldots, n$ there is

$$\binom{n}{x} \theta^x (1-\theta)^{n-x} \to \frac{e^{-\lambda}\lambda^x}{x!}$$

as $n \to \infty$ and $\theta = \frac{\lambda}{n}$.

This claim relates the "occurrence of rare events" nature of Poisson distribution.

Intuition: Consider a unit time period. Let $X \sim \mathrm{Poisson}(n\theta)$. Then X is the number of event occurrences during this period, with expectation $n\theta$.

On the other hand, partition the period to n equal intervals, and assume a chance θ that the rare event occurs in each interval. So the total number of events also follow $\mathrm{Binomial}(n, \theta)$. This total number is just X!

Remark.

- Requires θ close to 0, and n very large.
- If θ close to 1 and n very large, then $\mathrm{Binomial}(n, 1-\theta) \approx \mathrm{Poisson}(n(1-\theta))$.

Summary of discrete distributions:

Distribution	x values	pmf $p(x)$	$\mathbb{E}(X)$	$\mathrm{Var}(X)$
$\mathrm{Bernoulli}(\theta)$	$0, 1$	$\theta^x(1-\theta)^{1-x}$	θ	$\theta(1-\theta)$
$\mathrm{Binomial}(n, \theta)$	$0, 1, \ldots, n$	$\binom{n}{x}\theta^x(1-\theta)^{n-x}$	$n\theta$	$n\theta(1-\theta)$
$\mathrm{Geometric}(\theta)$	$1, 2, \ldots$	$(1-\theta)^{x-1}\theta$	$\frac{1}{\theta}$	$\frac{1-\theta}{\theta^2}$
$\mathrm{NB}(k, \theta)$	$k, k+1, \ldots$	$\binom{x-1}{k-1}(1-\theta)^{x-k}\theta^k$	$\frac{k}{\theta}$	$\frac{k\theta(1-\theta)}{\theta^2}$
$\mathrm{Poisson}(\lambda)$	$0, 1, \ldots$	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ